

Class 18: Online Learning and Concept Drift

Data-driven Systems Engineering

Agenda

- Why a validated model can still go stale.
- Offline versus online (incremental) learning.
- River: learning one observation at a time.
- Concept drift: what it is, and how ADWIN detects it.
- Monitoring a streaming model with River and Evidently together.

Learning goals

- Explain why some systems need incremental learning instead of periodic retraining.
- Use River's 'predict_one' / 'learn_one' pattern to reason about streaming models.
- Define concept drift and distinguish it from ordinary noise.
- Connect a drift detector such as ADWIN to a concrete monitoring and reset action.

Course Map and Running Cases



Today in the course

Focus: Learning continuously from streams, and detecting when the data itself has changed.

Bridge: Class 17 evaluated a fixed, trained model. This class asks what happens once that model has to keep working as new data keeps arriving.

Fraud case

The fraud case can use periodic batch retraining, but spending patterns and fraud tactics still drift over time between retraining cycles.

Pasteurization case

The pasteurization case is a natural streaming problem: sensors arrive continuously, and process or calibration changes must be detected as they happen.

Course thread: The course thread now shifts from a single validated snapshot to a model that must keep validating itself over time.

Offline versus online learning

Offline

- Train on a fixed dataset
- Easier benchmarking
- Periodic retraining

Online

- Update one sample at a time
- Works directly on streams
- Responds faster to change

Why adaptation becomes necessary

- User behavior changes.
- Sensors drift or get recalibrated.
- Business policies or regulations shift.
- New classes of errors appear only after deployment.

Key point

Model adaptation is not a sign that the first model failed. It is a recognition that the environment keeps evolving.

River example: incremental regression

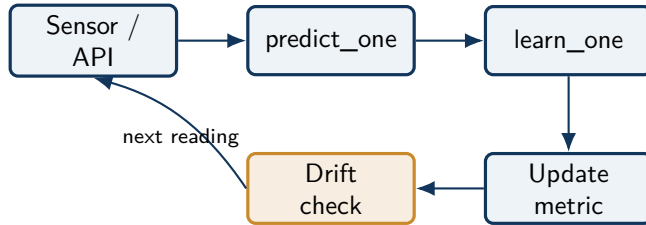
```
model = preprocessing.StandardScaler() | linear_model.LinearRegression(  
    optimizer=optim.SGD(0.01)  
)  
mae = metrics.MAE()  
  
y_pred = model.predict_one(x)  
model.learn_one(x, y)  
mae.update(y, y_pred)
```

- From 'notebooks/Class8_Model_Incremental_learning.ipynb': the model predicts, then learns, one observation at a time.
- There is no fixed training set; the model's parameters update continuously as data arrives from the Open-Meteo API.

What is concept drift?

- Concept drift is a change in the statistical relationship between inputs and target over time, not just noisy individual readings.
- Sudden drift: an abrupt regime change, such as a sensor recalibration or a new fraud pattern appearing overnight.
- Gradual drift: a slow shift, such as seasonal changes in spending or a wearing physical sensor.
- A model trained before the drift can keep looking confident while its predictions quietly become wrong.

The streaming learning loop



River example: ADWIN drift detection

```
adwin = drift.ADWIN(delta=0.002)
...
error = abs(y - (y_pred or 0))
adwin.update(error)

if adwin.drift_detected:
    print("Resetting model due to detected drift...")
    model = preprocessing.StandardScaler() | linear_model.LinearRegression(
        optimizer=optim.SGD(0.01)
    )
    adwin = drift.ADWIN(delta=0.002)
```

- From 'notebooks/Class8_Model_Incremental_learning.ipynb': ADWIN watches the prediction error stream and flags a statistically significant change.
- Detecting drift is only half the job; the notebook resets both the model and the detector once drift fires.

Why ADWIN and similar detectors matter

- ADWIN keeps an adaptive window of recent errors and compares its sub-windows for a statistically significant change.
- It grows the window while things look stable and shrinks it the moment the error distribution shifts.
- This gives a principled trigger for retraining or resetting, instead of a fixed, arbitrary retraining schedule.

From notebook to dashboard: the same pattern, deployed

```
y_pred = model.predict_one(x)
model.learn_one(x, y)
adwin.update(abs(temp - (y_pred or 0)))
drift_flag = adwin.drift_detected
```

- From 'streamlit/pages/App_Stream.py': the exact same River and ADWIN pattern from the notebook, now driving a live dashboard.
- Detected drift points are plotted directly against the mean absolute error curve, so an operator sees drift, not just a number.

Monitoring beyond a single detector

- 'notebooks/Class8_Model_Incremental_learning.ipynb' also integrates Evidently's 'DataDriftPreset' and 'DataQualityPreset' reports alongside River.
- A single statistical test (ADWIN on the error) is a fast, targeted signal; a full drift report checks many columns and quality dimensions at once.
- Combining a lightweight online detector with periodic, richer reports is a realistic production pattern, not an either-or choice.

Batch retraining versus online learning

Latency to adapt **Auditability**

Online reacts immediately; batch waits for the next retraining cycle.
Batch snapshots are easy to freeze and review; online state changes continuously.

Risk of instability **Infrastructure cost**

Online models can overreact to noise if the detector is too sensitive.
Online avoids large periodic retraining jobs but needs always-on serving.

Where drift shows up in the two running cases

- Fraud: seasonal spending shifts, new merchant categories, or fraud rings adopting a new pattern.
- Pasteurization: sensor recalibration, raw-material changes, or wear in the heating and cooling equipment.
- In both cases, drift detection turns “the model got worse” from a mystery into a specific, timestamped event.

Repo activity

- 'notebooks/Class8_Model_Incremental_learning.ipynb' is the primary anchor: River regression, ADWIN, and Evidently reports.
- 'streamlit/pages/App_Stream.py' deploys the same River and ADWIN pattern as a live dashboard.

Question

ADWIN just fired on the pasteurization temperature stream. Is that a sensor fault, a process change, or a false alarm, and how would you tell the difference?

- This connects drift detection back to the state-machine and signal reasoning from Class 13.

Papers and materials

River documentation; River ADWIN documentation; River metrics API; Gama et al., A survey on concept drift adaptation; Evidently on data drift detection.

From This Class to the Next

What stays with us

- Online learning updates a model one observation at a time, trading offline benchmarking convenience for responsiveness.
- Concept drift is a real, detectable statistical event, not a vague excuse for degraded performance.
- ADWIN-style detectors give a principled trigger for retraining or resetting a model.

Project use

Project teams should decide, for their own case, whether periodic retraining is enough or whether an online or drift-aware component is warranted.

Next connection

Class 19 is a hands-on lab: generating, serving, and streaming the pasteurization sensors that this class's drift ideas were grounded in.

Course thread: Detecting drift only matters if there is a running streaming system to detect it in, which is exactly what the next class builds.